Parser benchmark report: which parser is best, and what replaces Azure DI

The short answer

Two kinds of entries are compared. **Ready-made tools** are published parsers (Azure DI, pymupdf4llm, Docling and so on). **Our combos** (`hybrid_pdf`, `smart_pdf`) are written in this repo and glue existing pieces together: hybrid_pdf = PyMuPDF + Tesseract OCR; smart_pdf = PyMuPDF + Tesseract OCR + a vision model (Azure OpenAI) for figures. A combo winning means the combination works, not that a new parser exists.

- **Reference:** Azure Document Intelligence (ADI) scored **28.2** / **40** overall and is the best ready-made tool.
- **Best free ready-made alternative to ADI: pymupdf4llm** (27.9 / 40, 13.6 s per document). On mixed and scanned PDFs it is within 0.3 to 0.7 points of ADI.
- **Docling** (ready-made) matches ADI on quality (mixed 29.9, scanned 24.6) but is about 4 times slower than pymupdf4llm (56 s per document on average, 91 s on figure-heavy files).
- **Our combo hybrid_pdf** is the fastest option for digital PDFs and PDFs with tables (ties ADI: 25.0 on digital, 30.0 on a table PDF, 0.5 to 1.4 s). It is weak on scans (no headings).
- **Charts and diagrams: no ready-made parser reads them, including ADI** (Charts = 0). Only **our combo smart_pdf** did (Charts 4.5, Diagrams 5.0), because it adds a vision model. It costs one vision call per figure, so it is not a like-for-like comparison with the ready-made tools.
- **Not worth keeping:** LangChain, LlamaIndex and Haystack loaders (15.2: no OCR, no tables), OpenDataLoader (18.4: nothing on scans), pdfminer, pdfplumber, markitdown, pymupdf (cannot read scans at all).

Overall ranking, ready-made tools only

#	parser	Total /40	secs per document
1	**azure_di** (paid)	28.2	10.5
2	**pymupdf4llm**	27.9	13.6
3	**docling**	25.1	56.4
4	**easyocr**	21.2	199.7
5	**ocr** (Tesseract)	21.1	8.0
6	**pymupdf**	18.6	1.9
7	**opendataloader**	18.4	2.5
8	**markitdown, pdfplumber**	17.4	0.4 to 0.8
9	**langchain, llamaindex, haystack loaders**	15.2	0.2
10	**pdfminer**	15.1	0.3

Our combos

combo	built from	Total /40	secs per document
smart_pdf	PyMuPDF + Tesseract OCR + vision model	34.7	15.3
hybrid_pdf	PyMuPDF + Tesseract OCR	28.4	3.6

How the scoring works

Every parser ran on every PDF that has a hand-written answer key. Eight parameters, 0 to 5 each (same parameters as the Azure IDP vs OpenDataLoader email): Text, Tables, Charts, Diagrams (infographic), Numbers (entity and numeric fidelity), Structure, Complete, No-invent (5 = nothing made up). A parameter only counts when the document has something to test it with, so the total /40 is only comparable inside one table.

Overall (all documents)

parser	Text	Tables	Charts	Diagrams	Numbers	Structure	Complete	No-invent	Total /40	secs
smart_pdf (our combo: PyMuPDF + Tesseract + vision model)	4.3	4.9	4.5	5.0	3.8	3.3	4.9	4.0	**34.7**	15.3
hybrid_pdf (our combo: PyMuPDF + Tesseract)	4.6	4.2	0.0	2.5	4.3	3.5	4.8	4.5	**28.4**	3.6
azure_di	4.3	4.2	0.0	2.5	4.0	4.3	4.8	4.1	**28.2**	10.5
pymupdf4llm	4.2	4.2	0.0	2.5	4.2	4.0	4.8	4.0	**27.9**	13.6
docling	4.2	4.2	0.0	0.0	3.9	4.1	4.6	4.2	**25.1**	56.4
easyocr	4.2	0.0	0.0	2.5	3.6	2.0	4.7	4.1	**21.2**	199.7
ocr	4.3	0.0	0.0	2.5	3.4	2.1	4.7	4.1	**21.1**	8.0
pymupdf	3.2	2.1	0.0	0.6	3.3	2.7	3.3	3.3	**18.6**	1.9
opendataloader	3.2	2.1	0.0	0.6	3.3	2.7	3.2	3.3	**18.4**	2.5
markitdown	3.2	2.1	0.0	0.6	3.3	1.5	3.3	3.3	**17.4**	0.8
pdfplumber	3.2	2.1	0.0	0.6	3.3	1.6	3.3	3.3	**17.4**	0.4
langchain_loader	3.2	0.0	0.0	0.6	3.3	1.5	3.3	3.3	**15.2**	0.2
llamaindex_reader	3.2	0.0	0.0	0.6	3.3	1.5	3.3	3.3	**15.2**	0.2
haystack_converter	3.2	0.0	0.0	0.6	3.3	1.5	3.3	3.3	**15.2**	0.2
pdfminer	3.2	0.0	0.0	0.6	3.2	1.5	3.3	3.3	**15.1**	0.3

Digital PDF (1 document)

parser	Text	Tables	Charts	Diagrams	Numbers	Structure	Complete	No-invent	Total /40	secs
hybrid_pdf (our combo: PyMuPDF + Tesseract)	5.0	-	-	-	5.0	5.0	5.0	5.0	**25.0**	1.4
smart_pdf (our combo: PyMuPDF + Tesseract + vision model)	5.0	-	-	-	5.0	5.0	5.0	5.0	**25.0**	1.5

pymupdf	5.0	-	-	-	5.0	5.0	5.0	5.0	**25.0**	1.5
docling	5.0	-	-	-	5.0	5.0	5.0	5.0	**25.0**	27.6
opendataloader	5.0	-	-	-	5.0	4.9	5.0	5.0	**24.9**	2.2
azure_di	5.0	-	-	-	4.9	4.9	5.0	4.9	**24.7**	9.7
pymupdf4llm	4.8	-	-	-	4.9	4.8	5.0	4.6	**24.2**	2.2
pdfplumber	5.0	-	-	-	5.0	2.5	5.0	5.0	**22.5**	1.3
markitdown	5.0	-	-	-	5.0	2.5	5.0	5.0	**22.5**	2.1
pdfminer	5.0	-	-	-	5.0	2.5	5.0	5.0	**22.5**	0.8
langchain_loader	5.0	-	-	-	5.0	2.5	5.0	5.0	**22.5**	0.3
llamaindex_reader	5.0	-	-	-	5.0	2.5	5.0	5.0	**22.5**	0.2
haystack_converter	5.0	-	-	-	5.0	2.5	5.0	5.0	**22.5**	0.2
easyocr	4.9	-	-	-	4.6	2.5	4.9	4.8	**21.6**	201.2
ocr	4.9	-	-	-	3.7	2.5	4.9	4.9	**20.8**	8.4

Digital PDF with tables (1 document)

parser	Text	Tables	Charts	Diagrams	Numbers	Structure	Complete	No-invent	Total /40	secs
docling	5.0	5.0	-	-	5.0	5.0	5.0	5.0	**30.0**	10.4
opendataloader	5.0	5.0	-	-	5.0	5.0	5.0	5.0	**30.0**	1.7
pymupdf4llm	5.0	5.0	-	-	5.0	5.0	5.0	5.0	**30.0**	1.1
hybrid_pdf (our combo: PyMuPDF + Tesseract)	5.0	5.0	-	-	5.0	5.0	5.0	5.0	**30.0**	0.5
smart_pdf (our combo: PyMuPDF + Tesseract + vision model)	5.0	5.0	-	-	5.0	5.0	5.0	5.0	**30.0**	0.5
pymupdf	5.0	5.0	-	-	5.0	5.0	5.0	5.0	**30.0**	0.4
azure_di	5.0	5.0	-	-	5.0	5.0	5.0	5.0	**30.0**	6.9
markitdown	5.0	5.0	-	-	5.0	2.5	5.0	5.0	**27.5**	0.2
pdfplumber	5.0	5.0	-	-	5.0	2.5	5.0	5.0	**27.5**	0.0
easyocr	5.0	0.0	-	-	5.0	2.5	5.0	5.0	**22.5**	29.6
langchain_loader	5.0	0.0	-	-	5.0	2.5	5.0	5.0	**22.5**	0.0
llamaindex_reader	5.0	0.0	-	-	5.0	2.5	5.0	5.0	**22.5**	0.0
haystack_converter	5.0	0.0	-	-	5.0	2.5	5.0	5.0	**22.5**	0.0
pdfminer	5.0	0.0	-	-	5.0	2.2	5.0	5.0	**22.2**	0.0
ocr	4.9	0.0	-	-	4.8	2.5	4.9	4.9	**21.9**	1.0

Mixed PDF, digital plus scanned pages (2 documents)

parser	Text	Tables	Charts	Diagrams	Numbers	Structure	Complete	No-invent	Total /40	secs
docling	5.0	5.0	-	-	5.0	5.0	5.0	5.0	**29.9**	40.3
azure_di	5.0	5.0	-	-	5.0	4.9	5.0	5.0	**29.8**	11.9
pymupdf4llm	4.9	5.0	-	-	4.9	5.0	4.9	4.8	**29.5**	6.8
hybrid_pdf (our combo: PyMuPDF + Tesseract)	5.0	5.0	-	-	4.8	4.2	5.0	5.0	**28.9**	2.7
smart_pdf (our combo: PyMuPDF + Tesseract + vision model)	5.0	5.0	-	-	4.8	4.2	5.0	5.0	**28.9**	2.6
pymupdf	4.0	5.0	-	-	4.6	3.6	3.3	5.0	**25.4**	0.8
opendataloader	4.0	5.0	-	-	4.6	3.5	3.3	5.0	**25.3**	1.6
pdfplumber	4.0	5.0	-	-	4.6	1.9	3.3	5.0	**23.8**	0.3
markitdown	4.0	5.0	-	-	4.6	1.9	3.3	5.0	**23.8**	0.4
easyocr	4.9	0.0	-	-	4.8	2.5	4.9	4.9	**22.1**	130.6
ocr	4.9	0.0	-	-	4.2	2.5	4.9	4.9	**21.4**	4.6
langchain_loader	4.0	0.0	-	-	4.6	1.9	3.3	5.0	**18.8**	0.2
llamaindex_reader	4.0	0.0	-	-	4.6	1.9	3.3	5.0	**18.8**	0.2
haystack_converter	4.0	0.0	-	-	4.6	1.9	3.3	5.0	**18.8**	0.2
pdfminer	4.0	0.0	-	-	4.6	1.8	3.3	5.0	**18.7**	0.1

Scanned PDF (3 documents)

parser	Text	Tables	Charts	Diagrams	Numbers	Structure	Complete	No-invent	Total /40	secs
azure_di	5.0	-	-	-	4.9	5.0	5.0	5.0	**24.9**	9.8
docling	4.9	-	-	-	5.0	4.9	4.9	4.9	**24.6**	33.9
pymupdf4llm	4.9	-	-	-	4.9	4.7	4.8	4.9	**24.2**	18.2
easyocr	4.8	-	-	-	4.1	2.3	4.8	4.8	**20.9**	101.7
hybrid_pdf (our combo: PyMuPDF + Tesseract)	4.9	-	-	-	3.6	2.5	4.9	4.9	**20.6**	7.9
smart_pdf (our combo: PyMuPDF + Tesseract + vision model)	4.9	-	-	-	3.6	2.5	4.9	4.9	**20.6**	7.8
ocr	4.9	-	-	-	3.6	2.5	4.9	4.9	**20.6**	6.0
opendataloader	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	**0.0**	3.3
markitdown	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	**0.0**	0.1
pdfplumber	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	**0.0**	0.0
pdfminer	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	**0.0**	0.0

pymupdf	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	**0.0**	0.4
langchain_loader	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	**0.0**	0.0
llamaindex_reader	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	**0.0**	0.0
haystack_converter	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	**0.0**	0.0

PDFs with charts, figures, images and a tax table (5 documents)

parser	Text	Tables	Charts	Diagrams	Numbers	Structure	Complete	No-invent	Total /40	secs
smart_pdf (our combo: PyMuPDF + Tesseract + vision model)	3.4	4.6	4.5	5.0	3.0	2.9	4.8	2.7	**30.9**	30.8
hybrid_pdf (our combo: PyMuPDF + Tesseract)	4.2	2.5	0.0	2.5	4.0	3.1	4.6	4.0	**25.0**	2.6
pymupdf	4.2	2.5	0.0	2.5	4.0	3.1	4.6	4.0	**25.0**	3.7
opendataloader	4.0	2.5	0.0	2.5	4.0	3.1	4.4	4.0	**24.5**	2.5
markitdown	4.2	2.7	0.0	2.5	4.0	1.9	4.6	4.0	**23.8**	1.3
pdfplumber	4.1	2.6	0.0	2.5	4.0	2.0	4.6	3.9	**23.7**	0.7
azure_di	3.4	2.5	0.0	2.5	2.8	3.3	4.6	2.9	**22.0**	11.2
pymupdf4llm	3.3	2.5	0.0	2.5	3.3	2.9	4.8	2.7	**21.9**	18.3
langchain_loader	4.2	0.0	0.0	2.5	4.0	1.8	4.6	4.0	**21.0**	0.3
llamaindex_reader	4.2	0.0	0.0	2.5	4.0	1.8	4.6	4.0	**21.0**	0.3
haystack_converter	4.2	0.0	0.0	2.5	4.0	1.8	4.6	4.0	**21.0**	0.3
pdfminer	4.1	0.0	0.0	2.5	3.8	1.9	4.5	4.0	**20.9**	0.5
docling	3.3	2.5	0.0	0.0	2.5	3.0	4.0	3.1	**18.4**	91.4
ocr	3.4	0.0	0.0	2.5	2.8	1.5	4.4	3.1	**17.7**	11.7
easyocr	3.3	0.0	0.0	2.5	2.4	1.5	4.5	3.0	**17.2**	319.8

Which parser for which document

Document	Pick	Why	Free alternative to ADI
Digital text PDF	hybrid_pdf (our combo)	ties ADI (25.0 vs 24.7), 1.4 s	pymupdf (same score, 1.5 s)
PDF with many tables	hybrid_pdf or pymupdf4llm	30.0, same as ADI, under 1.5 s	pymupdf4llm (keeps Markdown tables)
Mixed (digital + scanned)	pymupdf4llm	29.5 vs ADI 29.8, 6.8 s	docling (29.9) if time does not matter
Scanned PDF	pymupdf4llm	24.2 vs ADI 24.9, 18 s	docling (24.6, 34 s)
Charts, diagrams, infographics	smart_pdf (our combo, needs a vision model)	only option that reads values and arrows	none free without a vision model

Limits of this test (read before deciding)

- **The test set is small:** 1 digital, 1 table, 2 mixed, 3 scanned and 5 figure documents. Treat small gaps (under about 1 point) as ties.
- **Charts and diagrams are scored on a single PDF** (`chart_sample.pdf`). Their result is a first look, not a proof.
- **smart_pdf's "No-invent" score drops on figure-heavy documents** (2.7) because the vision model adds descriptions that are not in the answer key. That is partly a scoring artefact, not necessarily errors.
- **Not tested:** LlamaParse (API key empty), Unstructured (crashes on import on this machine), MinerU and Marker (not installed), Databricks ai_parse_document (paid, no access).
- **ADI was re-scored** after fixing the scorer so that HTML tags are not counted as words; the other rows come from the first run (their output has almost no HTML).
- **Speed** was measured on one Windows laptop and is only comparable inside this report.

Large documents (no answer key: closeness to Azure DI, so not absolute accuracy)

Run 1: three PDFs of 24, 39 and 11 pages (74 pages). Run 2: Llama 2 paper (77 pages) and UN SDG Report 2024 (51 pages). Totals are out of 40 against Azure DI's own output; Charts, Diagrams and Tables are left out because they cannot be measured this way.

Parser	Run 1 total	Run 1 s/page	Run 2 total	Run 2 s/page
pymupdf4llm	23.5	2.5	26.6	2.9
hybrid_pdf (our combo: PyMuPDF + Tesseract)	22.7	1.5	24.4	1.1
pymupdf	22.5	1.5	24.4	1.7
smart_pdf (our combo: PyMuPDF + Tesseract + vision model)	21.7	2.7	24.2	2.8
opendataloader	22.4	0.3	11.7 (Java ran out of memory on the UN report)	0.5
docling	20.6	16.8	24.9	6.5 on the UN report only
markitdown	20.8	0.5	18.4	1.1
pdfplumber	20.7	0.4	16.7	1.1
azure_di	reference	0.8	reference	0.7

Docling's Llama 2 time is discarded: the laptop went to sleep during that run and the clock read 8 hours. The full report with charts and diagrams is the Claude doc "Free alternatives to Azure Document Intelligence".